

UGED1111 Logic

Basic Concepts I

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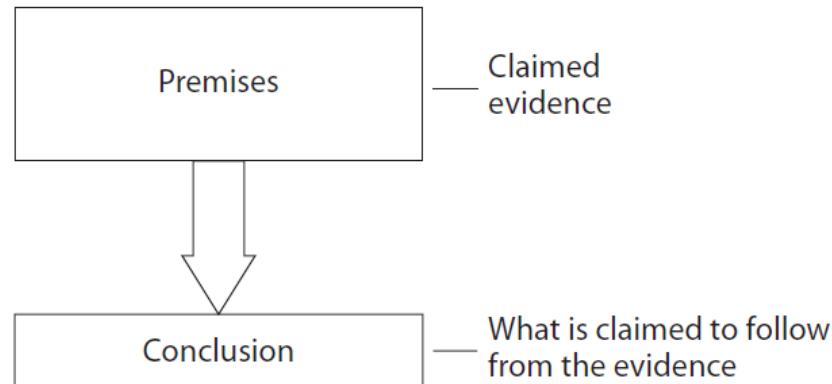
Agenda



- **Basic elements and structure of an argument**
- **How to recognize an argument**
- **Deductive argument and Inductive argument**

Basic Form of an Argument

- ◆ The **premises** are the **statement(s)** that set forth the **reasons or evidence**.
- ◆ The **conclusion** is the **statement** that the evidence is claimed to support/ justify.



Statement

- Premise(s) and Conclusion must be **statements**
- **Declarative sentence** that is ***either true or false***
- Grammatical

Example:

1. Today is 17th May 2024.
2. 韓國瑜是韓國人
3. Water is H₂O.

Discussion 1

Are they statements?

1. How is today's weather?
2. Snow white slow Is
3. 如果全球氣溫再上升五度，香港就會消失。
4. 球波一係入、一係唔入。
5. Leave me alone!
6. $2+2=2$

Fact check?

與您同賀美心60周年·免費送紀念版月餅

**不要再傳了！
是2016年的！**

只要香港身份證號碼有
"60"或"18"兩組相連數字，
抑或身份證姓名有
"美"或"心"字

到美心月餅 Pop-up Store
免費獲贈「美心60周年紀念
版月餅」1盒(價值\$248)，
每日限量100盒！

8月13日-9月4日期間的星期六、日



求驗傳媒

· 8月8日 ·

年份的重要性

美心又送月餅？這幾天Whatsapp流傳的圖，大家傳得不亦樂乎，但大家可會覺得該圖有點似曾相識呢？

其實，那圖美心是2016年的「60周年」優惠，而美心不是譚詠麟，不會年年都「60歲」。在2017年，也有網民把這個2016年的優惠誤傳，結果美心也有澄清。

今年已經是2019年，謠言之可以無限復活，最主要原因是該圖沒有年份，也沒有網站資料供參考。

所以，還是那一句，任何沒有年月日及資料來源的資訊，大家都要小心處理，不要輕易轉發。

2016年的新聞：

<https://www.weekendhk.com/462909/dining/美心月餅-免費-60周年紀念版/>

Truth

Truth is the attribute of those statement that assert what really is the case (Copi, pp26)

****Corresponding to the fact?***

Example:

- 1. Today is Monday.**
- 2. There is a cat on the Bed.**
- 3. There is no female in this class.**



Standard Form of an Argument

Premise₁

Premise₂

Premise₃

...

Premise_n $(n \geq 1)$

Conclusion

Typical Example



1. Tom is a CUHK Students. (Premise)
 2. All CUHK Students are smart.(Premise)
-

3. Tom is smart. (Conclusion)

Premise Indicators

Typical premise indicators are

■ Since ; in that; seeing that; as indicated by; may be inferred from; for the reason that; because; as in as much as; for given that; owing to; etc

■ 由於、基於、理由是、因為; etc

Example: 由於GPA 高可以得到獎學金，我一定要好好讀書。

Premise: GPA 高可以得到獎學金。

Conclusion: 我一定要好好讀書。

Conclusion Indicators

Typical conclusion indicators are

■ Therefore; accordingly; entails that; wherefore we may conclude; hence; thus; it must be that ;it follows that; consequently; for this reason; implies that; etc

■ 因此、所以、故此、結論是, etc

Example:只有信耶穌才得永生，所以你應該信耶穌。

Premise:只有信耶穌才得永生。

Conclusion:你應該信耶穌

Hidden Premise

Hidden Premise: A premise which assumed in the argument but not stated explicitly

Example:

只有信耶穌才得永生，所以你應該信耶穌。

Recap



- **Statement – either true or false**
- **We will not say an argument is “true” or “false”.**
- ***** “statement” and “proposition”?**

Question

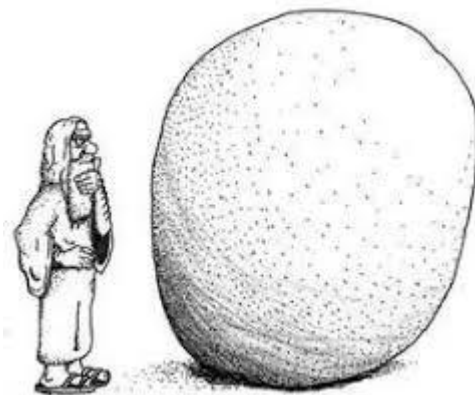
Is this a statement?



Example

石頭論證

如果上帝無法造出自己舉不起來的石頭，那上帝不是全能的。如果上帝可以造出自己舉不起來的石頭，上帝也不是全能的。要不上帝可以造出自己舉不起來的石頭，要不上帝無法造出自己舉不起來的石頭。所以上帝不是全能的。

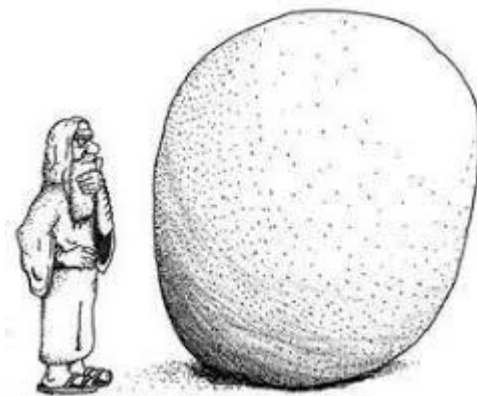


Example - standard form

Standard form

1. 如果上帝無法造出自己舉不起來的石頭，上帝不是全能的。
2. 如果上帝可以造出自己舉不起來的石頭，上帝不是全能的。
3. 上帝可以造出自己舉不起來的石頭或上帝無法造出自己舉不起來的石頭。

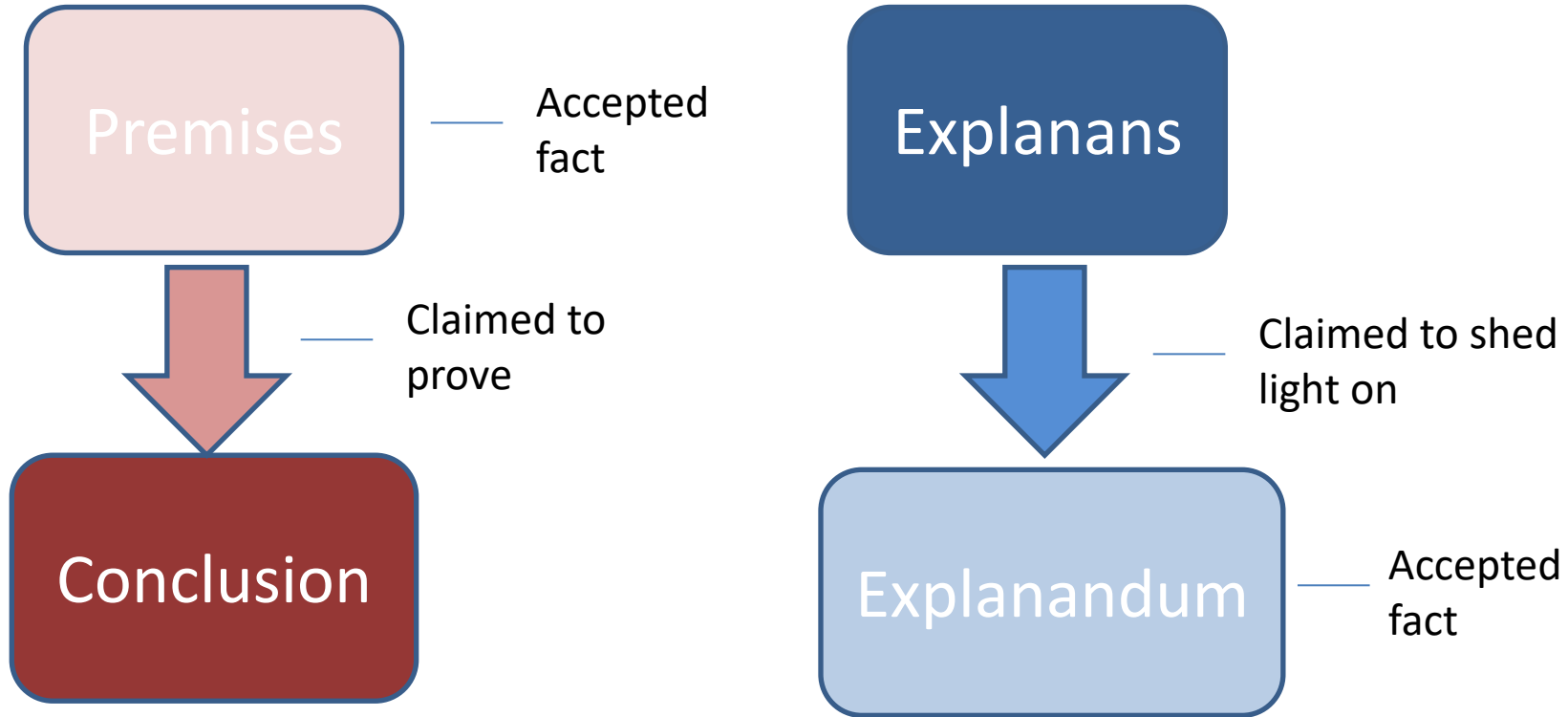
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4. 上帝不是全能的。



NOT -Argument

- ❑ A Single statement
- ❑ Conditional sentence
(e.g. if it is raining, Tom will bring an umbrella.)
- ❑ A set of statements not intend to support any claim
- ❑ Advice / Warning/ Illustrations, etc
- ❑ Causal explanation

Argument Vs Explanation



Explanation



Explanans: statement(s) which is claimed to give the explanation.

Explanandum: statement which describes the accepted fact.

Argument Vs Explanation

□ Example 1:

這隻狗有跳蚤，因為牠身上有在蠕動的黑色點，而且你還可以看見牠不停在抓癢。

□ Example 2

這隻狗有跳蚤，因為昨天帶牠去公園時牠和野狗一起玩，還在草地滾來滾去。

Reference

- Copi, I., Cohen, C., & McMahon, K. (2011). *Introduction to Logic (14th ed.)*. Upper Saddle River, NJ: Pearson Education.
- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.
- Lau, J. (2011). *An Introduction to Critical Thinking and Creativity : Think More, Think Better*. Hoboken, N.J.: Wiley.